

Menopause Management in Patients With Cancer

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This Narrative Review provides a practical, symptom-based framework for the management of menopause in patients with cancer, emphasizing risk stratification and collaboration with oncology. Current evidence and guideline recommendations were synthesized from randomized trials, observational studies, systematic reviews, and major society position statements, with findings organized by symptom domain and stratified by cancer type and hormone sensitivity to support clinical decision making. Menopausal symptoms are common and often severe in patients undergoing cancer treatment, particularly in those with treatment-induced menopause, and management requires balancing symptom relief with oncologic safety. Nonhormonal therapies are first-line treatment across cancer types, with strong evidence supporting selective serotonin and norepinephrine reuptake inhibitors, gabapentin, and lifestyle interventions for vasomotor symptoms. Vaginal estrogen is effective for genitourinary syndrome of menopause, is appropriate in most gynecologic cancers, and may be considered in select breast cancer survivors after shared decision making. Systemic hormone therapy remains contraindicated in hormone receptor–positive breast cancer but may be considered in carefully selected patients with non–

hormone-dependent malignancies or low-risk gynecologic cancers. Nonhormonal considerations—including thrombotic risk, hepatic function, drug–drug interactions, and cardiometabolic health—are critical to safe management. A symptom-based, risk-stratified approach enables gynecologists to manage menopause in patients with cancer through collaboration with oncology, supporting individualized care and improved quality of life. (*Obstet Gynecol* 2026;00:1–11)

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The incidence of breast and gynecologic cancers continues to rise in the United States,¹ exposing an increasing number of patients to treatments that induce menopause. Therapies, including cytotoxic chemotherapy, ovarian suppression, bilateral oophorectomy, pelvic radiation, and endocrine agents such as aromatase inhibitors, are essential for reducing recurrence and improving survival. However, these treatments frequently result in ovarian failure and menopause, leading to vasomotor symptoms, genitourinary syndrome of menopause (GSM), sexual dysfunction, sleep disturbance, and mood changes. Symptom onset is often abrupt and severe, occurring during a period already marked by psychological and physical stress.²

Despite the prevalence and significance of treatment-induced menopause, patients with cancer consistently report dissatisfaction with anticipatory guidance and symptom management.³ Many describe inadequate counseling regarding expected effects on quality of life and limited access to effective therapies.^{4,5} Untreated menopausal symptoms are associated with decreased adherence to endocrine therapies, particularly in hormone receptor–positive breast cancer, which may increase recurrence risk and negatively affect survival.^{6,7}

Oncologists are highly skilled in delivering therapies that reduce tumor burden and recurrence risk; however, menopause management is not a primary focus of oncology training.^{8,9} Although the physiologic effects of treatment-induced menopause are recognized,

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evolving therapies and competing clinical demands limit familiarity with management strategies. Many oncologists report discomfort managing menopausal symptoms and uncertainty about the safety of hormone therapy (HT) in cancer survivors.^{2,10} Even among gynecologic oncologists, practice patterns vary, and evidence-based therapies are not consistently used.¹¹

Conversely, gynecologic specialists have expertise in menopause management but often report discomfort caring for patients with a history of cancer, particularly those with hormone-dependent malignancies.¹² This bidirectional knowledge gap contributes to fragmented care, persistent symptoms, and reduced quality of life. Gynecologists are therefore uniquely positioned to play a central role in menopause management of cancer survivors through collaborative care with oncology.

This review synthesizes current evidence on the management of menopausal symptoms in patients with cancer, with emphasis on hormone-dependent breast and gynecologic cancers. We focus on practical, evidence-based strategies to support gynecologists in safely co-managing these patients with oncology, improving quality of life and supporting adherence to oncologic therapies (Box 1).

METHODS

This review synthesizes current evidence on the management of menopausal symptoms in patients

Box 1. Framework for an Integrated Clinical Approach for the Management of Menopausal Symptoms in Patients With Cancer

1. Assess cancer type and hormone sensitivity
 - a. High risk: hormone receptor–positive breast, leiomyosarcoma or endometrial stromal sarcoma, granulosa cell tumors
 - b. Moderate risk: hormone receptor–positive breast, high-risk endometrial, low grade serous ovarian
2. Start with nonhormonal therapies
3. Escalate to local therapies (eg, vaginal estrogen) when appropriate
4. Consider systemic hormone therapy only in carefully selected patients
5. Collaborate with oncology
 - a. Consider hormone-containing therapies (systemic or local) in patients with a high-risk or moderate-risk cancer type
 - b. Consider systemic hormone therapy in patients with increased thromboembolic risk (active tumor, history of deep vein thrombosis)
 - c. Consider therapy that may impair hepatic function or metabolism in patients on active or maintenance treatments

with breast and gynecologic cancers, with a focus on hormone-dependent malignancies. Open Evidence AI was queried to identify recent reviews and meta-analyses, followed by a structured PubMed search of studies published from January 2000 through March 2026. Search terms included combinations of menopause-related symptoms, cancer types, and treatment strategies, including hormonal and nonhormonal therapies.

Eligible studies included randomized controlled trials, observational studies, systematic reviews, meta-analyses, and clinical practice guidelines evaluating the safety and efficacy of menopause therapies in patients with a history of cancer. Emphasis was placed on hormone receptor–positive breast cancer and estrogen-sensitive gynecologic malignancies. Studies in noncancer populations were included when cancer-specific data were limited but interpreted with caution.

Guidelines and consensus statements from major organizations were reviewed, including those from the North American Menopause Society, the American College of Obstetricians & Gynecologists, the American Society of Clinical Oncology, the National Comprehensive Cancer Network (NCCN), the European Society of Endocrinology, the International Menopause Society, and the Society of Gynecologic Oncology.^{13–19} Reference lists of selected articles were also screened for additional relevant studies. Given the heterogeneity in cancer types and treatments, findings were synthesized narratively and stratified by cancer type and hormone receptor status to support shared decision making and co-management between gynecology and oncology.

Mechanisms of Treatment-Induced Menopause in Patients With Cancer

The average age of menopause in developed countries is approximately 51 years,²⁰ but menopause occurs earlier in patients undergoing cancer treatment at a mean age of 44 years.²¹ Earlier menopause is associated with increased risks of cardiovascular disease, osteoporosis, and other chronic conditions.²² Cancer-directed therapies induce menopause through multiple mechanisms, often resulting in abrupt hypoestrogenism. Unlike the gradual transition of natural menopause, treatment-induced menopause is typically acute and associated with more severe, disruptive symptoms that significantly impair quality of life.^{3,23–25}

Bilateral oophorectomy, commonly performed for gynecologic and pelvic malignancies, results in immediate loss of ovarian hormone production. It

may also be used in patients with hormone receptor-positive breast cancer or prophylactically in patients with hereditary cancer syndromes.^{3,26} Surgical menopause is associated with more severe vasomotor symptoms, sexual dysfunction, mood disturbance, and accelerated bone loss compared with natural menopause.²³

Pelvic radiation is highly gonadotoxic, leading to ovarian failure in up to 90–100% of patients, depending on age and dose. It is frequently used in cervical, endometrial, colorectal, and other pelvic malignancies, placing a broad population at risk.²⁷ Cytotoxic chemotherapy similarly causes ovarian dysfunction through follicular damage, with risk varying by agent, dose, and patient age. Alkylating agents carry the highest risk of permanent ovarian failure.²⁸

Endocrine therapies for hormone receptor-positive breast cancer further contribute to menopausal symptoms. Tamoxifen acts as a selective estrogen receptor modulator, whereas aromatase inhibitors suppress peripheral estrogen production and require concurrent ovarian suppression in premenopausal patients.^{7,29,30} Aromatase inhibitors induce more profound estrogen deprivation than tamoxifen, resulting in more severe vasomotor symptoms and genitourinary atrophy. Limited observational data suggest that menopausal symptoms are more frequent, more severe, and more persistent after breast cancer treatment than after natural menopause.^{23,24}

Hematopoietic stem cell transplantation is another important cause of treatment-induced menopause, with ovarian failure rates exceeding 90% as a result of high-dose chemotherapy and total body irradiation. Symptoms may be further exacerbated by graft-versus-host disease and immunosuppressive therapy.³¹

Collectively, these treatments contribute to the high prevalence and severity of menopausal symptoms in cancer survivors. The abrupt onset of hypoestrogenism, combined with the stress of cancer diagnosis and treatment, often results in symptom profiles more severe than those of natural menopause. Recognition of these mechanisms is essential to guide anticipatory counseling and early collaborative management.

Hormonal Status in Breast and Gynecologic Cancers

Hormone receptor status is central to oncologic treatment and the safety of menopausal HT. In breast and select gynecologic cancers, tumor growth may be influenced by estrogen and progesterone, making assessment of hormonal sensitivity essential.

Specific recommendations for and against HT vary substantially by cancer subgroup and are supported by distinct evidence bases. For hormone receptor-positive breast cancer, the HABITS (Hormonal Replacement Therapy After Breast Cancer—Is It Safe?) trial demonstrated a hazard ratio of 3.3 for cancer recurrence in survivors randomized to HT compared with control, and meta-analyses confirm this risk; systemic HT is therefore contraindicated.^{32–34} For hormone receptor-negative (triple-negative) breast cancer, the biological concern is lower, but evidence is insufficient to recommend systemic HT.³⁵ For early-stage, low-grade endometrioid endometrial cancer, a randomized Gynecologic Oncology Group trial and multiple meta-analyses have not demonstrated increased recurrence with HT, supporting its consideration in selected patients.^{36–39} For high-risk or advanced-stage endometrial cancer, evidence is sparse and significant caution is required.^{39–42} For epithelial ovarian cancer (high-grade serous carcinoma), the AHT (Adjuvant Hormone Therapy) randomized trial showed no harm and a possible survival benefit with HT.^{43–46} These distinctions are summarized in Table 1.

Approximately 70–80% of breast cancers are estrogen receptor-positive.⁴⁷ Estrogen promotes proliferation in these tumors, and reducing estrogen exposure improves recurrence-free and overall survival.⁴⁸ Standard therapies include selective estrogen receptor modulators such as tamoxifen and aromatase inhibitors, which suppress peripheral estrogen production.^{7,29} In premenopausal patients, these are combined with ovarian suppression or oophorectomy.³⁰

Progesterone receptor (PR) status provides additional prognostic value, with PR-positive tumors indicating intact estrogen signaling and greater responsiveness to endocrine therapy, whereas estrogen receptor-positive/PR-negative tumors may exhibit more aggressive biology.⁴⁹ Human epidermal growth factor receptor 2 status refines prognosis but does not directly alter menopause management decisions.⁵⁰

Triple-negative breast cancers lack estrogen receptor, PR, and human epidermal growth factor receptor 2 expression and are not hormonally driven.⁵⁰ Although endocrine therapies are ineffective, caution is warranted when systemic HT is considered because of limited safety data.

Among gynecologic cancers, hormonal sensitivity varies by histology. Endometrial cancer is the most clearly estrogen dependent, particularly low-grade endometrioid tumors, which may respond to progestin

therapy.⁵¹ High-grade endometrial cancers are generally not hormonally driven.⁵²

Ovarian cancer is heterogeneous. High-grade serous tumors are not strongly hormone dependent, whereas low-grade serous and endometrioid subtypes may express hormone receptors and respond to endocrine therapies.^{53,54} Cervical, vaginal, and vulvar cancers are not hormone dependent, and menopausal HT is generally considered safer in this population. Before HT is prescribed, individual risk factors must be considered.⁵⁵

Symptom-Based Management of Menopause in Patients With Cancer and Survivors

Management of menopausal symptoms in patients with cancer should follow a symptom-based, risk-adapted approach that integrates symptom severity with tumor biology. The NCCN recommends screening for symptoms that affect quality of life, although specific intervals are not defined. Validated tools such as the Menopause-Specific Quality of Life Questionnaire and Menopause Rating Scale have been studied in oncology populations,⁵⁶ but simple screening questions are often sufficient in clinical practice.¹⁶

Because the safety of hormonal therapies depends on tumor hormone sensitivity, management must align symptom burden with oncologic risk. Nonhormonal therapies are first-line treatment for hormone-dependent cancers, whereas local and, in select cases, systemic hormonal therapies may be considered in lower-risk or non-hormone-dependent malignancies. Multidisciplinary collaboration and shared decision making are essential (Tables 1 and 2).^{2,16}

Vasomotor Symptoms

Vasomotor symptoms are common and disruptive. Management generally begins with nonhormonal therapies. In breast cancer, particularly hormone receptor-positive disease, neurokinin-targeted therapies and selective serotonin reuptake inhibitors/serotonin-norepinephrine reuptake inhibitors are recommended first-line treatments per updated NCCN guidelines. Elinzanetant, a dual neurokinin-1 and -3 receptor antagonist, is approved in Europe for the treatment of vasomotor symptoms resulting from endocrine therapy in breast cancer survivors and has demonstrated significant improvement in vasomotor symptoms and sleep in clinical trials.⁵⁷ Fezolinetant, a selective neurokinin-3 receptor antagonist,^{58–60} has also demonstrated efficacy; a clinical trial in breast cancer survivors is in progress (NCT06917313). Selective serotonin reuptake inhibitors/serotonin-norepinephrine reuptake inhibitors (eg, venlafaxine,

citalopram) and gabapentin remain effective alternatives; strong CYP2D6 inhibitors (eg, paroxetine, fluoxetine) should be avoided in patients receiving tamoxifen. Oxybutynin may also be considered. Systemic HT is generally avoided.^{2,61,62}

Young breast cancer survivors with treatment-induced premature ovarian insufficiency face long-term health consequences from hypoestrogenism, including accelerated bone loss, cardiovascular disease, and cognitive effects. Shared decision making about the potential role and timing of HT should be incorporated into survivorship planning, particularly for those with early-stage, non-hormone receptor-positive disease.⁶³ *BRCA* carriers who undergo risk-reducing salpingo-oophorectomy without a prior breast cancer diagnosis represent a distinct group; observational data support HT use until the average age of natural menopause (approximately 51 years) because the health consequences of premature surgical menopause outweigh the theoretical hormonal risk in the absence of breast cancer.^{2,63} Racial and ethnic differences in symptom severity should be considered: Black women with cancer experience more frequent and severe vasomotor symptoms, underscoring the importance of individualized, culturally responsive assessment.⁶⁴

In gynecologic cancers, management depends on tumor type and risk. Nonhormonal therapies are first-line treatment for most patients, although systemic HT may be considered in selected patients with severe symptoms after oncologic consultation. Hormone-sensitive tumors favor nonhormonal approaches, whereas systemic HT may be considered in non-hormone-dependent cancers such as cervical and vulvar malignancies.^{19,65} Overall, nonhormonal therapy should be initiated across cancer types, with escalation to systemic HT reserved for carefully selected patients when oncologic safety has been established.

Genitourinary Syndrome of Menopause

Genitourinary syndrome of menopause, including vaginal dryness, dyspareunia, and urinary symptoms, is common and often undertreated in cancer survivors. Initial management should focus on nonhormonal therapies, including vaginal moisturizers and lubricants. Patients treated with aromatase inhibitors experience particularly severe genitourinary atrophy resulting from profound estrogen deprivation, often requiring earlier escalation to vaginal estrogen therapy.^{12,14}

In patients with breast cancer, low-dose vaginal estrogen may be considered when symptoms persist and significantly affect quality of life. This is generally more acceptable in patients receiving tamoxifen;

Table 1. General Recommendations for Systemic Hormone Therapy in Patients With Cancer and Survivors

Cancer Type	Systemic HT	Key Considerations
Breast		
HR positive (ER/PR+)	Avoid	Increased recurrence in HABITS trial; meta-analysis confirms risk in HR+ disease ^{32–34}
HR negative	Rare	Limited data; lower biological concern but insufficient safety evidence ^{2,20,35}
Endometrial or uterine		
Endometrial–early stage	Reasonable	Meta-analyses show no increased recurrence in early-stage disease ^{36–39}
Endometrial–high risk or advanced	Caution	Sparse data; consider tumor biology and receptor status ^{39–42}
Uterine sarcoma (LMS, ESS)	Avoid	Hormone-sensitive tumors; endocrine therapy used in recurrence ⁴¹
Ovarian		
Epithelial (high-grade serous carcinoma)	Reasonable	RCT shows no harm and possible survival benefit (AHT trial) ^{43–46}
Low-grade serous or endometrioid	Caution	Hormone-responsive subtypes; endocrine therapy used clinically ⁴¹
Granulosa cell	Avoid	Strong estrogen sensitivity; avoid exogenous hormones ⁴¹
Cervical	Reasonable	Not hormone driven; standard menopause approach ⁴¹
Vulvar or vaginal	Reasonable	Not hormone driven ⁴¹
Colon	Reasonable	Observational data show no increased mortality; possible benefit
Lung	Reasonable; see considerations	Estrogen-only therapy carries no apparent increased risk; combined estrogen–progesterone therapy warrants more caution and should be individualized; caution in smokers ^{2,16}
Hematologic malignancy	Reasonable	Not hormone driven; consider VTE risk and transplantation history ²
Post-HCT (premature ovarian insufficiency)	Recommended	Strong indication for HT in young patients with POI after transplantation ²

HT, hormone therapy; HR, hormone receptor; ER, estrogen receptor; PR, progesterone receptor; HABITS, Hormonal Replacement Therapy After Breast Cancer–Is It Safe?; LMS, leiomyosarcoma; ESS, endometrial stromal sarcoma; RCT, randomized controlled trial; AHT, Adjuvant Hormone Therapy; VTE, venous thromboembolism; HCT, Hematopoietic cell transplantation; POI, premature ovarian insufficiency.

greater caution is warranted in those on aromatase inhibitors, and use should involve shared decision making with oncology.^{14,66}

Recent evidence suggests that vaginal estrogen is not associated with increased breast cancer recurrence or mortality overall.⁶⁷ However, in a longitudinal national cohort study, a subgroup analysis of patients receiving aromatase inhibitors, representing approximately 5% of participants, demonstrated a potential increase in recurrence risk without a corresponding increase in mortality, and its safety in this population remains uncertain.⁶⁸

Aromatase inhibitors work by inhibiting the aromatase enzyme, substantially reducing systemic estrogen production, but do not directly block estrogen receptor activity at the tissue level.⁶⁹ Any systemically absorbed estrogen from vaginal preparations may therefore stimulate estrogen receptor–positive breast tissue. Available data demonstrate minimal systemic absorption, with most studies showing either transient or no significant increases in serum estradiol levels.⁷⁰ Switching from an aromatase inhibitor to tamoxifen may be considered in select patients

because tamoxifen blocks estrogen at the receptor level.⁷¹ The NCCN guidelines note that vaginal estrogen preparations with lower systemic absorption such as rings or suppositories are preferred over creams, although further safety data are needed, particularly for patients on aromatase inhibitors.¹⁶

In contrast, vaginal estrogen is generally safe and effective in patients with gynecologic cancers, including low-risk endometrial and epithelial ovarian cancers, as well as cervical, vulvar, and vaginal cancers. A more cautious, individualized approach is recommended for hormone-sensitive tumors, as outlined in the British Gynaecological Cancer Society and British Menopause Society guidelines.⁶⁵ Overall, vaginal estrogen is an effective therapy for GSM and should not be unnecessarily withheld, particularly in non-hormone-dependent cancers.

Sexual Dysfunction

Sexual dysfunction in cancer survivors is multifactorial and often underrecognized, requiring a multimodal approach addressing both physiologic and psychosocial factors.^{2,16}

Table 2. Recommended Therapy for Menopausal Symptoms in Patients With Cancer and Survivors

Symptom	1st-Line Therapy	2nd-Line Therapy	When to Use Hormones
Vasomotor	SSRIs or SNRIs (venlafaxine), gabapentin, neurokinin-targeted therapy (elinzanetant)	Oxybutynin; fezolinetant (HIGHLIGHT 1 ongoing, NCT06440967)	Systemic HT if cancer type allows
GSM	Vaginal moisturizers or lubricants	Pelvic floor PT	Vaginal estrogen (preferred over systemic) Rarely systemic HT alone
Sexual dysfunction	Lubricants, pelvic floor PT, counseling	Vaginal estrogen	
Musculoskeletal pain	Exercise intervention	Acupuncture, CBT	Systemic HT if appropriate
Sleep	Control vasomotor symptoms, gabapentin, CBT	Short-term sedative-hypnotics	Systemic HT if appropriate
Mood	SSRIs or SNRIs	CBT	Systemic HT may be beneficial
Bone loss	Calcium or vitamin D, weight-bearing exercise	Bisphosphonates or denosumab	Systemic HT if appropriate
Cardiometabolic health	Lifestyle modification and treatment of risk factors		HT not recommended for cardiometabolic risk factors

SSRI, selective serotonin reuptake inhibitors; SNRI, serotonin–norepinephrine reuptake inhibitor; HIGHLIGHT 1, A Study to Confirm if Fezolinetant Helps Reduce Hot Flashes in Women With Breast Cancer Who Are Having Hormone Therapy; HT, hormone therapy; GSM, genitourinary syndrome of menopause; PT, physical therapy; CBT, cognitive behavioral therapy.

Management across cancer types should include optimization of GSM, use of vaginal lubricants, and referral for pelvic floor physical therapy when indicated. Vaginal dilator therapy is particularly important after pelvic radiation, with evidence supporting improved function with regular use.⁷² Psychosexual counseling should also be considered.¹⁶

Pharmacologic options are limited. Nonhormonal agents such as bupropion or flibanserin may be considered in select patients, although evidence is modest. Vaginal estrogen improves dyspareunia and tissue health in most non–hormone-dependent cancers and may be used selectively in breast cancer survivors after counseling.^{14,16}

Intravaginal dehydroepiandrosterone (prasterone) is metabolized locally to androgens and estrogens with limited systemic absorption, providing a rationale for its use even in some hormone-sensitive patients; topical testosterone similarly exerts predominantly local effects. Both are included in NCCN 2026 survivorship guidelines as options for vaginal symptoms.¹⁶ Evidence in hormonally mediated cancers remains limited, and oncologic consultation is advisable before use. Topical androgens may also be considered for hypoactive sexual desire disorder, although data are limited.

Musculoskeletal Syndrome of Menopause

Musculoskeletal symptoms affect 57–71% of menopausal women and are common in cancer survivors.^{73–76}

Evidence supports nonhormonal interventions, particularly exercise-based approaches such as resistance training, high-intensity exercise, and yoga.^{77,78} Acupuncture has shown benefit in breast cancer populations.⁵ Cognitive behavioral therapy and mindfulness may be helpful according to broader menopause and chronic pain data.^{79,80}

Sleep Disturbance

Sleep disturbance is common and often multifactorial.¹⁶ Management should focus on underlying contributors, particularly vasomotor symptoms. Nonhormonal therapies, including gabapentin, may improve nocturnal symptoms, and cognitive behavioral therapy for insomnia should be considered when available.^{2,16} Elinzanetant has also demonstrated sleep benefit in clinical trials.⁵⁷ Short-term sedative-hypnotics may be used cautiously.^{2,16}

Mood Changes

Mood disturbances, including anxiety and depression, are common in treatment-induced menopause and are often biologically mediated. Selective serotonin reuptake inhibitors and serotonin–norepinephrine reuptake inhibitors are first-line therapies and can also improve vasomotor symptoms.^{2,16} In patients receiving tamoxifen, strong CYP2D6 inhibitors should be avoided.² Otherwise, management is similar across cancer types.

Bone Loss and Fracture Risk

Premature estrogen deprivation leads to accelerated bone loss and increased fracture risk, particularly in patients receiving aromatase inhibitors or ovarian suppression.^{3,81} Management includes calcium and vitamin D supplementation, weight-bearing exercise, and routine bone density monitoring. Pharmacologic therapy with bisphosphonates or denosumab should be used when indicated.⁸¹

Bisphosphonates have also demonstrated oncologic benefit. In a meta-analysis of individual patient data from 11,767 postmenopausal women in randomized trials, bisphosphonates were associated with significant reductions in recurrence and breast cancer mortality, with no benefit observed in premenopausal patients.⁸²

Cardiometabolic Health

Premature menopause is associated with increased cardiovascular risk, insulin resistance, and adverse lipid profiles.³ Management should focus on lifestyle modification and treatment of cardiometabolic risk factors, including hypertension, dyslipidemia, and glucose abnormalities.¹⁶

Nonhormonal Considerations in the Management of Menopause in Patients With Cancer

In addition to hormone receptor status, the management of menopausal symptoms in patients with cancer requires consideration of nonhormonal factors that influence treatment safety and efficacy. Cancer and its treatment are associated with physiologic changes, including hypercoagulability, hepatic dysfunction, and altered drug metabolism, that affect therapeutic decision making (Table 3).

Hypercoagulability is common, with patients with cancer demonstrating a fourfold to sevenfold increased risk of venous thromboembolism.⁸³ This risk is further increased by therapies such as chemotherapy, antiangiogenic agents, and tamoxifen.^{84,85} Systemic menopausal HT, particularly oral estrogen, may further elevate thrombotic risk and is generally avoided in patients with active malignancy or high baseline risk. Transdermal estrogen may offer a lower-risk alternative in selected patients.⁸⁶

Hepatic function is another key consideration; many oncologic and menopause therapies are metabolized through cytochrome P450 pathways.⁸⁷ Tamoxifen requires CYP2D6 metabolism to its active form, and concurrent use of CYP2D6 inhibitors (eg, paroxetine, fluoxetine) may reduce efficacy and should be avoided.^{62,88} Alternatives such as venlafaxine, citalo-

pram, or gabapentin are preferred.⁸⁸ Hepatic impairment may also increase drug exposure and toxicity, requiring dose adjustment and close monitoring.⁶⁷ Rare liver injury has been reported with fezolinetant; elinzanetant did not demonstrate hepatotoxicity in clinical trials.⁸⁹

Cancer therapies also affect cardiometabolic health. Aromatase inhibitors are associated with adverse lipid profiles, bone loss, and increased cardiovascular risk.^{90,91} Similarly, treatment-induced menopause increases the risk of osteoporosis, sarcopenia, and cardiometabolic disease.⁹² Management should therefore include bone health monitoring, supplementation, exercise, and cardiovascular risk reduction.

Polypharmacy is common in cancer survivors, increasing the risk of drug interactions and cumulative adverse effects. Many nonhormonal therapies have side effects that overlap with cancer-related symptoms, necessitating individualized treatment selection and coordination with oncology.⁹³ Overall, effective management requires a comprehensive approach that incorporates thrombotic risk, hepatic function, drug interactions, and long-term metabolic health.

CONCLUSIONS

The management of menopause in patients with cancer requires a deliberate, individualized approach grounded in risk stratification and shared decision making. Risk is not uniform across cancer types: tumor hormone sensitivity, treatment exposures, comorbid conditions, and patient-specific factors all influence the safety profile of available therapies. Treatment decisions should therefore be guided by structured assessment of oncologic risk and symptom burden rather than blanket avoidance.

Shared decision making is essential. Patients must be actively engaged in discussions about the risks, benefits, and uncertainties of treatment options, with particular attention to how symptoms affect daily function and quality of life.

Collaboration between gynecology and oncology is equally critical. Nonhormonal therapies remain first-line treatment across all cancer types and should be initiated early. However, appropriately selected patients, particularly those with lower-risk or non-hormone-dependent malignancies, may benefit from hormonal therapies when decisions are made within a collaborative, risk-informed framework. The goal is to optimize quality of life without compromising oncologic outcomes.

Menopause management in patients with cancer should be viewed as a dynamic process that evolves

Table 3. Nonhormonal Considerations in the Management of Menopause in Patients With Cancer and Survivors

Consideration	Clinical Issue	Considerations for Menopause Management	Recommended Approach
Hypercoagulability and VTE risk	Cancer increases VTE risk 4–7 times; further increased by chemotherapy, antiangiogenic agents, and tamoxifen	Systemic estrogen (especially oral) further increases thrombotic risk	Avoid systemic HT in active patients and patients at high risk; consider transdermal estrogen if HT is necessary
Drug–drug interactions (CYP pathways)	Many oncology drugs metabolized via CYP450; tamoxifen requires CYP2D6 activation	CYP2D6 inhibitors reduce tamoxifen efficacy	Avoid paroxetine, fluoxetine; use venlafaxine, citalopram, escitalopram, or gabapentin
Hepatic function	Liver dysfunction from cancer or therapy alters drug metabolism	Increased drug exposure and toxicity (hormonal+nonhormonal)	Monitor LFTs; adjust dosing; avoid hepatically intensive agents when possible
Cardiometabolic risk	Cancer therapies (AIs, chemotherapy) worsen lipid profiles, cardiovascular risk	Limits systemic HT use; increases long-term morbidity	Emphasize lifestyle, lipid control, BP management; consider nonhormonal therapy
Bone health	Estrogen deprivation (AIs, oophorectomy) accelerates bone loss	Increased fracture risk; HT often contraindicated	DEXA screening, calcium or vitamin D, weight-bearing exercise, bisphosphonates or denosumab
Premature menopause	Treatment-induced ovarian failure→early estrogen deficiency	Increased long-term risk (cardiovascular, bone, cognitive)	Consider HT in appropriate cancers; aggressive risk factor modification if not
Polypharmacy	Multiple concurrent medications common in patients with cancer	Increased adverse effects, drug interactions, symptom overlap	Choose agents with minimal interactions; simplify regimens; monitor closely

VTE, venous thromboembolism; HT, hormone therapy; LFT, liver function test; AI, aromatase inhibitor; BP, blood pressure; DEXA, dual-energy X-ray absorptiometry.

with disease trajectory. By prioritizing risk stratification, interdisciplinary communication, and patient engagement, clinicians can deliver care that is both evidence based and responsive to individual patient needs.

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